

A Tuning Transformer for the Automatic Adjustment of Resonant Loop Antennas in RFID Systems

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Abstract—Radio Frequency Identification (RFID) systems allow contactless communication between a small portable tag and a reader device. In this paper, we address the problem of performance degradations due to the detuning of resonant reader antennas in the widely used 13.56 MHz frequency band. A novel design of a tuning transformer for the adjustment of the center frequency of resonant loop antennas is proposed. The ferrite core of the transformer is premagnetized using an additional DC winding. The HF and DC windings are decoupled by an iron core. Besides the tuning, the transformer may also be used for impedance matching and for balancing an unbalanced signal. The function of the device was proven by network measurements.

Key Words: Loop antenna, resonance tuning, RFID, tunable inductance.

1 Introduction

Systems for Radio Frequency Identification (RFID) allow contactless communication with portable tags. Such tags consist of a microelectronic circuit with memory and an antenna. The data on the tag can be read and frequently also be written by a reader device. The reader features an antenna, allowing the non-contacting data transfer between tag and reader, a transmitter and a receiver. In passive RFID systems, the power for the operation of the tag is supplied through the field of the reader antenna. This allows the production of very cheap, battery-free and therefore compact tags. The operating frequencies of RFID systems range from 135 kHz up to the GHz band. Many commercial systems use a center frequency of 13.56 MHz, which is also the target frequency used in this research project. Reading ranges of up to 80 cm with a card-size tag and

a single antenna have been reported with 13.56 MHz devices (e.g. [1]).

RFID systems are well established in a wide range of applications. They are, e.g., beneficially used for anti-theft protection in department stores, in the public transport sector, for electronic ticketing, animal identification and access control systems [2]. RFID solutions are also used in industrial automation and logistics. Still, there are numerous emerging industrial applications, making the RFID business a fast-growing and dynamic market. RFID systems for applications in supply-chain management, part identification, tool identification and quality control may help to increase the level of automation and the flexibility of production processes.

Especially in industrial environments, performance degradations of RFID systems may occur due to the presence of moving field-distorting objects, like vehicles, machinery and workpieces. Such objects influence the antenna efficiency, resulting in a reduced reading range. A novel tuning transformer is presented in this paper to compensate for the effects of such disturbances of the RFID reader performance.

In the following section the operating principle of a 13.56 MHz reader and the circuit including the proposed tuning transformer are described. After that, the tuning device itself is discussed in detail. Finally, experimental results are given to demonstrate its applicability to resonance tuning.

2 Tunable Reader Operating Principle

In 13.56 MHz RFID systems inductive coupling is used for the transmission of power and data. Both reader and transponder are therefore equipped with loop antennas. A power amplifier is used in the reader to generate the strong magnetic field required to power the transponders within the operating range. For optimal power transmission between reader and tag, the

reader antenna is tuned to resonance at the operating frequency using an additional capacitor. If a transponder is placed within the magnetic field of the reader, the reader and tag antennas act as a transformer with a very low coupling. In that way the transponder is supplied with energy from the reader field. For the communication from the reader to the tag, it is common to use amplitude modulation of the carrier (e.g. [3]). The transponder tag responds with a modulation of the load resistance connected to the tag antenna/resonance circuit, causing a change of the reader field and consequently of the induced voltage in the reader antenna. The voltage signal is tapped and demodulated to recover the data sent by the tag.

The tuning circuit of the reader antenna is usually equipped with a damping resistor in order to limit the quality factor Q of the resonant circuit. This step is necessary to let the modulation sidebands from load modulation pass through the antenna circuit. In addition, component tolerances and drift of component values with temperature and time cause deviations from the nominal resonant frequency. The effective inductance of the loop antenna itself depends on the operating environment in which the reader is placed. Metals or other electrically active materials in the vicinity of the antenna influence both its inductance and resistance. Especially industrial environments may be quite disturbing and varying in a wide range. A common way of accounting for impedance variations is to lower the quality factor Q of the resonant circuit. Then the RFID system can even operate with a detuned resonant circuit. The disadvantage of this approach is decreased transmission efficiency, reducing the operating range of the reader system.

An alternative is to use one or more tunable components to readjust the resonant frequency of the reader circuit, as proposed in [4, 5]. The tuning can be achieved with a set of binarily weighted, switched capacitances that are addressed by a microcontroller [5]. However, this method only allows for discontinuous adjustment of the resonant frequency. A continuously variable capacitor is used in [6] for the tuning of an amateur radio loop antenna. The capacitor consists of two overlapping coaxial tubes that can be displaced electromechanically by a DC drive. For RFID applications such an assembly is impractical because of the installation size and the low achievable tuning speed.

Therefore a novel device for the automatic tuning of resonant antennas is proposed. It consists of a tunable inductance that can be used to compensate for changes of the antenna inductance. In addition, it may be used for impedance transformation and for balancing the antenna. A series resonant circuit is used in the driving circuit of the antenna [2]. The tuning transformer is included as series inductance with the loop antenna connected to its secondary, as depicted in Fig. 1. The antenna is modeled as an R-L parallel circuit. Its self-capacitance only amounts to a few pF and is therefore

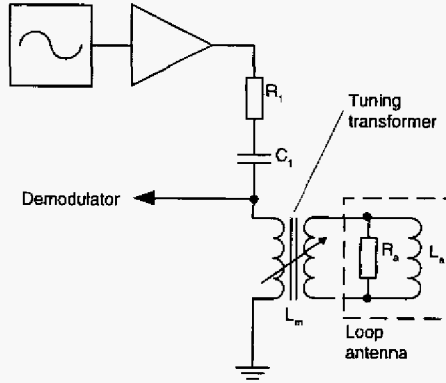


Fig.1: Antenna driving circuit of a 13.56 MHz RFID reader with a saturable tuning transformer. The resonant frequency is tuned by changing the magnetizing inductance of the transformer.

disregarded. The resonant frequency of the driving circuit depends on the magnetizing inductance of the tuning transformer.

The adjustable range of the resonant frequency depends on the ratio of magnetizing and antenna inductance. The smaller the magnetizing inductance, the bigger is the adjustable range, but at the expense of an increased magnetization current. The load modulated carrier signal is tapped at the primary winding of the transformer for demodulation.

Assuming a transformer with a perfect coupling and a magnetizing inductance L_m , the resonant frequency of the circuit in Fig. 1 is:

$$f_r = \frac{(n^2 L_a + L_m) R_a}{2\pi \sqrt{L_m n^4 L_a^2 R_a^2 C_1 + L_m^2 n^2 L_a R_a^2 C_1 - L_m^2 L_a^2}} \quad (1)$$

Thus, the resonance can be shifted by tuning L_m . In Fig. 2 the curve of the resonant frequency is plotted against the magnetizing inductance for a transformer ratio of $n = 2.5$. The horizontal line indicates the target frequency of 13.56 MHz. Values of $L_a = 2\mu\text{H}$ and $R_a = 2.1\text{k}\Omega$ obtained from a rectangular 50x50 cm antenna

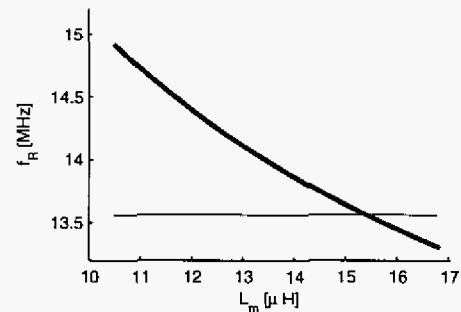


Fig.2: Resonant frequency of the reader circuit as a function of the tuning transformer magnetizing inductance.

are used, and C_1 is set to 20pF.

3 Tuning Transformer

The magnetizing inductance of the proposed tuning transformer can be continuously adjusted in a certain range. This allows for the automatic tuning of resonant circuits. The magnetizing inductance is tuned by applying a DC flux to the transformer core. This principle has long been used to implement variable inductances [7]. Designs for high frequency resonant power converters are proposed in [8, 9]. All approaches have in common that the DC flux is generated by DC windings located on the same core as the AC winding. AC voltages are therefore induced in the DC windings. Even when two DC windings are connected in series to cancel out the induced AC voltages, an AC voltage remains due to imperfections in the structure of the device [8]. A solution for the VHF range is described in [10]. A stripline inductor is tuned with the help of an orthogonal DC field through planar ferrite slices. However, this method is not applicable for the lower frequencies of the present RFID application.

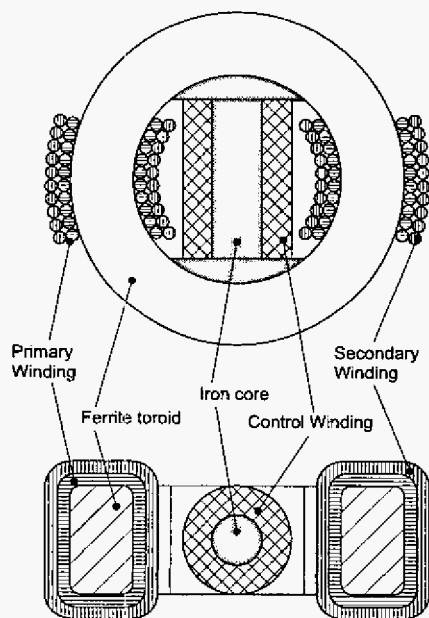


Fig.3: Principle of the tuning transformer with primary and secondary HF windings on a ferrite toroid and the DC control winding on an iron core.

The tunable transformer proposed in this paper uses a different approach for decoupling the AC and DC windings. The structure of the device is shown in Fig. 3. The main part is a standard ferrite toroidal core suited for high frequency applications. The primary and secondary windings both consist of two opposite parts that are connected in series. The DC field is generated by a control winding placed in the center of the ferrite core. The field is guided in a core made of standard transformer iron. The DC flux path is closed through the two

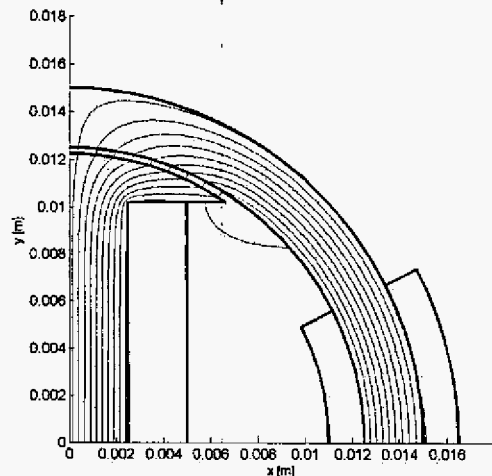


Fig.4: Magnetic flux lines in the tuning transformer with DC excitation in the control winding. The whole field is fed into the ferrite core.

halves of the toroid. By applying a DC control current, the ferrite core can be partly saturated, changing the permeability of the core. The HF signal is superimposed on the premagnetization. The amplitudes used in RFID systems result in small flux densities of a few mT which can be disregarded compared with the DC magnetization. Fig. 4 shows a finite element simulation of the DC magnetic field distribution in the tuning transformer. Only one quarter of the device is shown. The DC field is generated by a current flowing in the control winding. The entire magnetic flux is coupled into the ferrite core with an initial relative permeability of $\mu_r = 125$. The DC field can thus be used to saturate the ferrite material along the whole circumference of the ring core. This results in a change of the magnetizing inductance of the transformer. Because of the high conductivity of the iron, the high frequency magnetic field cannot penetrate the iron core and is kept in the toroid. Eddy current losses in the iron core are negligible because of the small skin depth of a few μm . The HF and DC windings are very effectively decoupled with the proposed method. This is illustrated in Fig. 5. The simulation shows the magnetic field distribution in the tuning transformer for a 13.56 MHz excitation current in the primary windings. Almost the whole field is concentrated in the ferrite material. The iron core does not distort the field because the eddy currents induced at the surface of the iron core block the magnetic field. For comparison, the same simulation is performed with a 50 Hz excitation current in Fig. 6. The magnetic field deeply enters the iron and a part of the magnetic flux path is even closed through the iron core. This flux causes excessive eddy current losses. Only when the excitation frequency is increased to above a few kHz, the iron core blocks the magnetic field and the tuning transformer can be used as intended.

Fig. 7 illustrates the measured variation of the inductance of the primary side with open secondary wind-

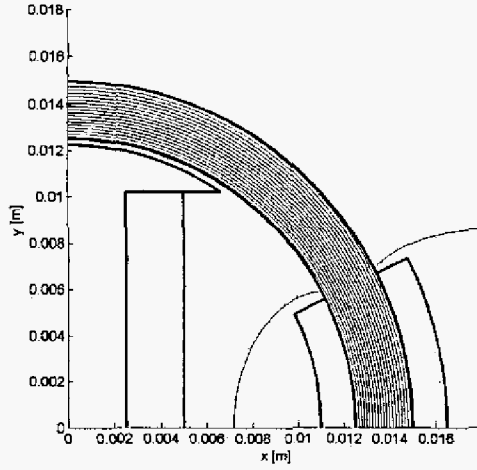


Fig.5: Magnetic flux lines in the tuning transformer with 13.56 MHz excitation in the primary windings. The magnetic field cannot penetrate the iron core.

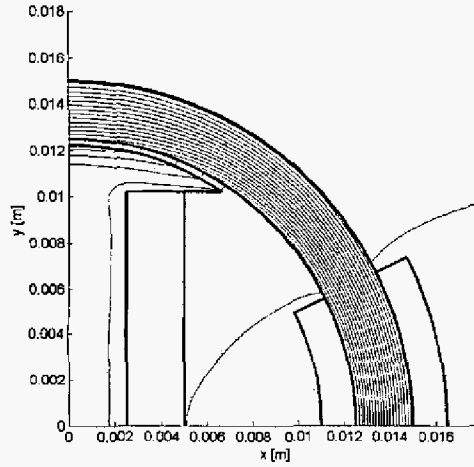


Fig.6: Magnetic flux lines in the tuning transformer with 50 Hz excitation in the primary windings. The magnetic field penetrates the iron core, leading to excessive losses.

ings as a function of the DC control current. The inductance remains constant up to a control current of 60 mA, but continuously falls off at higher currents. The maximum DC current of 300 mA results in a power dissipation of 2 W in the control coil. If the measured inductance $L_m = f(I_{DC})$ is inserted in (1), the expected resonant frequency of the reader circuit is obtained as a function of the DC current (see Fig. 8). The frequency remains constant at low currents because the magnetizing inductance of the transformer is constant, but steadily increases above about 60 mA. An adjustable range of 1.6 MHz can be reached with a current variation of 300 mA.

The saturable tuning transformer also offers additional functionality in the same device. It can be used for impedance transformation by choosing an appropriate transformation ratio. The balancing of an unbal-

anced signal can be achieved by grounding the middle tap of the secondary winding.

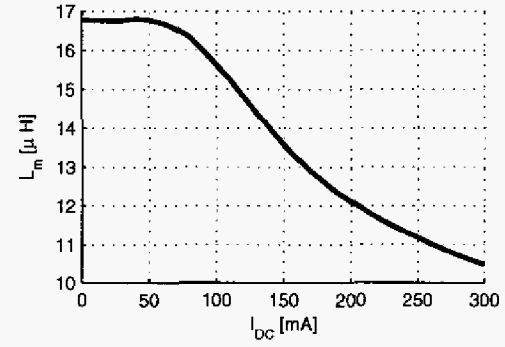


Fig.7: Measured primary open loop inductance of the tuning transformer as a function of the DC control current.

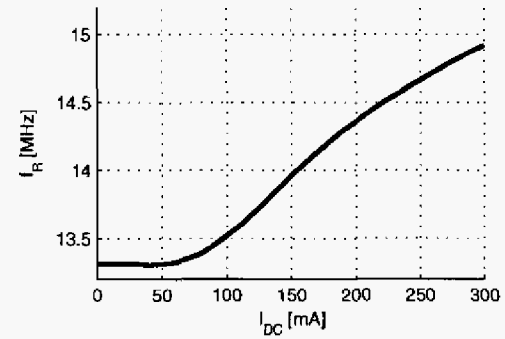


Fig.8: Calculated resonant frequency of the reader circuit as a function of the DC control current.

4 Experimental Results

Experimental results were obtained from a prototype built according to Fig. 3 with a transformer ratio of $n = 2.5$. A ferrite core with an outer diameter of 36 mm was used. The impedance of the resonant circuit was tuned to be approximately 50Ω at 13.56 MHz and a quality factor Q of about 20 was used, a typical value as a compromise between a strong magnetic field and a low damping of the modulation sidebands [2].

Fig. 9 shows the effect of the premagnetizing current on the center frequency of the resonant circuit as measured with a network analyzer. Due to the nonlinear characteristic of the permeability of the ferrite material the inductance decreases with increasing premagnetization and the center frequency increases as well. The bandwidth of the circuit does not change significantly. For practical applications a center frequency of 13.56 MHz should occur with medium premagnetization. Typically, conductive materials (e.g. a car), near the antenna lead to a decrease of the effective antenna inductance, which can be compensated for with lower

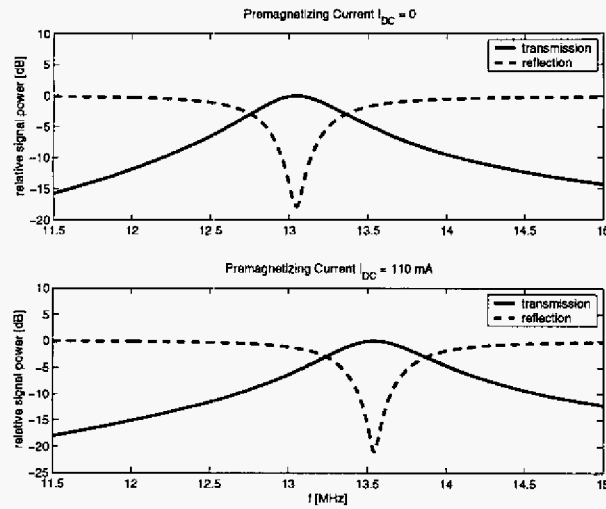


Fig.9: Resonant circuit characteristic for different premagnetizing currents. Due to premagnetization, the center frequency increases.

preomagnetization. Since detuning also affects the phase of the antenna current, this can be easily detected and corrected by the reader controller.

Fig. 10 shows the measured dependence of the center frequency on the DC premagnetization. The curve shows excellent agreement with the expected values shown in Fig. 8. However, the value of the capacitor C_1 of the reader circuit had to be lowered to 10 pF due to additional parasitic capacitances. For the used ferrite material the nonlinear change of the permeability starts at a premagnetization current of about 60 mA. The maximum current that can be used depends on the maximum power dissipation of the DC coil. For our prototype, with a maximum control power of 2 W (corresponding to $I_{DC} = 300$ mA), a frequency shift of 1.6 MHz is achieved. The relation between premagnetization current and center frequency is only slightly nonlinear in the range from 70 mA to 300 mA and therefore well suited for automatic control. The quality factor Q of the resonant circuit slightly decreases at the higher frequencies due to the lowered reactance.

5 Conclusion

A novel design of a tuning transformer for the adjustment of the center frequency of resonant loop antennas in RFID reader systems has been proposed. The change of inductance is caused by a premagnetization of the ferrite core of the transformer due to the nonlinear magnetization curve of the material. A control winding is used to generate the DC field for premagnetization. The decoupling between the HF and DC windings is achieved by using an iron core for the control coil. Besides the tuning, the transformer may also be used for impedance matching and for the balancing of an unbalanced signal. Due to the promising experimental results the use of the tuning device for other applications and

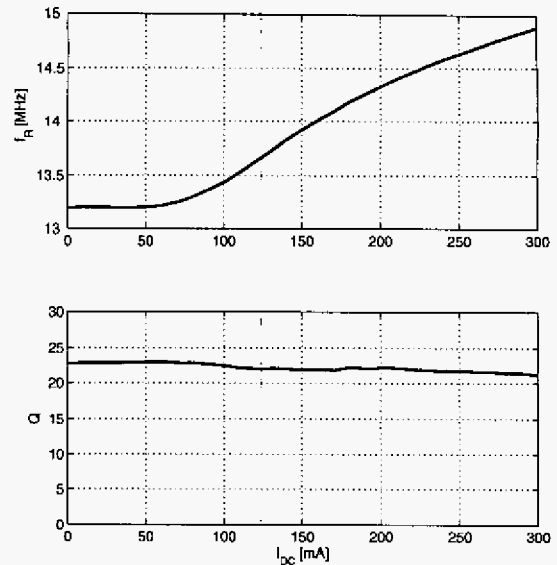


Fig.10: Center frequency and quality factor of the resonant circuit as a function of the premagnetizing current I_{DC} .

higher frequencies will be investigated.

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